

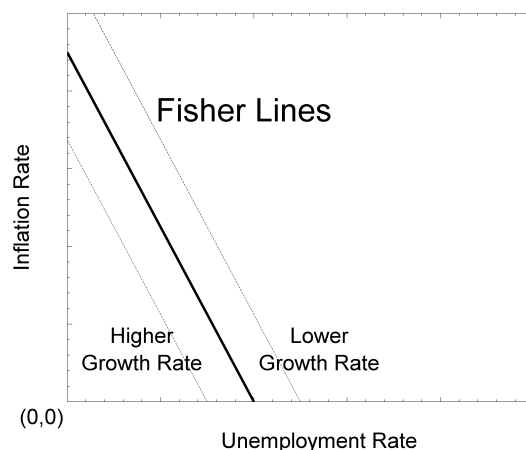
ENTROPIC ECONOMICS AND FINANCIAL CRISES

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The negative feedback response of prices and quantities to excess supply and demand is at the core of economic theory. Current macroeconomic theory explains unemployment as a restriction which prevents this negative feedback process from reaching equilibrium, however there is no clear understanding of the actual mechanism or good empirical evidence to support this theory. We examine the macroeconomic theory of the physicist Henri Rathgeber (1908-1995). Hypothesising that the aggregate agreements of buyers and sellers is reduced by an error rate to determine the actual set of transactions, we can use Shannon's information theory to find a physically determined constraint on the unemployment rate and inflation rate as shown in the figure.



This condition is not stable as there exists a positive feedback cycle between increases in the inflation rate and growth that pushes the Fisher line to the right. This fundamental solution leads to a broad range of applications. For instance we can isolate the effects of financial crises. Typically a reduction in the inflation rate causes an increase in the unemployment rate as determined by a movement along the Fisher line. Examining unemployment rate and inflation rate data confirms this relationship but also reveals some 'unusual' behaviours that can only be explained as rapid exogenous changes in the growth rate. Particularly striking examples of these occur in Korea between 1997 and 2001 and in Finland between 1990 and 1994. Both these countries experienced financial crises during these periods. Liquidity traps such as in the United Kingdom between 1921 and 1938 and in Japan between 1992 and 2007 are another cause of significant changes in the growth rate.

Keywords

information theory, unemployment, inflation, macroeconomics, econophysics